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Open trailer assembly

Abstract

An open trailer for carrying an all-terrain vehicle (ATV) has a front hoop assemble that is slidably connected to tongue assembly and a rear hoop assembly that that is fixedly connected to a proximal end of the tongue assembly. The rear hoop assembly comprises an axle to which are attached to tires and two ramps. When in use, the vehicle is driven up the ramps and onto the trailer until the front wheels of the vehicle are supported by the front hoop assembly. As the vehicle continues to be driven onto the trailer, the front wheels push the front hoop assembly toward a distal end of the tongue assembly until the rear wheel of the vehicle are supported by the rear hoop assembly.

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Background/Summary

1. FIELD OF THE INVENTION

(1) The present disclosure relates to an open trailer design suitable for carrying a vehicle from one place to another.

2. BACKGROUND

(2) Open trailers suitable for general-purpose use are available in many different sizes and can be used to carry a wide variety of vehicles of different dimensions, functionality and weights. Unfortunately, any particular general-purpose trailer design may not be ideally suited to carry all types of vehicles, and so special purpose trailers have been designed to carry particular types of vehicles. For example, a large, open trailer designed to carry cars may not be practical for carrying smaller vehicles, such as motorcycles or ATVs, as it may be too expensive, too large to tow behind some vehicles, and not easily configured to accept an ATV for travel. Consequently, open trailers have been designed with characteristics (i.e., dimensions, configuration, durability/strength, etc.) that are suitable to carry particular types of vehicles.

(3) Open trailers designed for carrying cars are typically larger, stronger, and costlier than trailers designed to carry All-Terrain Vehicles (ATVs), Utility-Terrain Vehicles (UTV) or Side-by-Side (SxS) Vehicles, all of which will be referred to hereinafter as ATVs. On the other hand, open trailers suitable for carrying smaller vehicles, such as an ATV, are typically designed with lighter weight, less expensive materials, a single axle with relatively small tires, one or more removable ramps used to safely drive the ATV onto the trailer, and some mechanism with which to secure or tie-down the vehicle while the trailer is being towed.

(4) While many larger trailers, such as car carriers, are typically stored out-of-doors, it may be desirable to store a smaller, ATV trailer in a garage or shed, or some other covered structure. In this regard, some smaller trailers are designed to be collapsible for storage in a smaller space, such as a garage or shed.

(5) Further, open trailers are typically designed with a bed (i.e., surface upon which a vehicle rests) having a particular length and width that is suitable to the particular type of vehicle being towed. So, for example, a car carrier bed length and width should be designed with dimensions sufficient to carry cars having a variety of dimensions. Still further, and depending upon the application, it may be desirable for the trailer bed to not be comprised of a solid material, but rather a metallic, mesh material that is lighter in weight and allows debris (i.e., dirt or grass material) to fall through the bed. Still further, it may be desirable for a trailer to not have a bed at all, but rather have two tracks, each having a particular width and length, and being appropriately spaced to allow a two or four wheeled vehicle to be driven onto the trailer.

Description

3. BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an illustration of a trailer assembly **10**

(2) FIG. 2A is an illustration showing an axle hoop assembly **11** comprising the trailer assembly **10**.

(3) FIG. 2B is an illustration showing the axle hoop assembly **11** exploded to show component parts.

(4) FIG. 3A is an illustration showing a tongue assembly **12** comprising the trailer assembly **10**.

(5) FIG. 3B is an illustration showing the tongue assembly **12** exploded to show component parts.

(6) FIG. 4A is an illustration showing a front hoop assembly **13** comprising the trailer assembly **10**.

(7) FIG. 4B is an illustration showing the front hoop assembly **13** exploded to show component parts.

(8) FIG. 5A is an illustration showing a front hoop carriage assembly **14** comprising the trailer assembly **10** with a portion of the tongue assembly **12**.

(9) FIG. 5B is another illustration showing the front hoop carriage assembly **14** without the portion of the tongue assembly **12**.

(10) FIG. 5C is an illustration of the front hoop carriage assembly **14** showing bearings **56A** and **56B**.

(11) FIG. 6 is an illustration of a bottom slider wheel **54**.

4. DETAILED DESCRIPTION

(12) Typically, open trailers designed to carry relatively small vehicles, such as ATVs, have a bed comprised of a single sheet of material, and the single sheet can either be composed of a plywood or a metal mesh material. While an open trailer having a bed comprised of a single sheet of material can be capable of carrying a wide variety of small vehicles, they tend to be heavy and not designed to be disassembled or easily collapsed for storage, if they are collapsible at all. The weight of such open trailers can be attributed to the relatively large dimensions of the bed necessary to carry ATVs of different sizes. In this regard, a bed comprising a general-purpose ATV trailer has to be long enough and wide enough to carry the range of ATVs.

(13) We have designed a general-purpose open trailer suitable for carrying ATVs having a wide range of wheel bases and axle lengths, and that can be easily disassembled into its component parts for storage. This trailer does not have a bed comprised of a single sheet of material, is light-weight, and can easily be configured or re-configured to carry ATVs having different wheel bases and axle length.

(14) According to one embodiment, our trailer design has a first or axle hoop assembly and a second or front hoop assembly both of which function to support and to retain rear and front wheels

of an ATV respectively. The axle hoop assembly is fixedly connected to a tongue assembly comprising the trailer, and the front hoop assembly is slidably connected to the tongue assembly and serves to receive front wheels of an ATV as it is driven onto the trailer. Once the ATV front wheels have engaged with the front hoop assembly, it can be easily moved from a starting position, at a proximal or axle end of the tongue assembly, to a final position towards a distal or hitch ball end of the tongue assembly as the ATV is driven onto the trailer. This movement of the front hoop assembly along the length of the tongue assembly, while an ATV is driven onto the trailer, effectively permits the trailer assembly to be dynamically configured (self-configured) to accept an ATV of any wheel base without operator intervention while the ATV is driven onto the trailer.

(15) According to another embodiment, the axle hoop assembly is comprised of first and second axle tube sliders that slide into and out of opposite ends of a first and a second axle tube that are fixedly connected to the proximal end of the tongue assembly. Further, the front hoop assembly is comprised of first and second front hoop sliders that slide into and out of opposite ends of a first front hoop tube and a second front hoop tube both of which are fixedly connected to the distal hitch ball end of the tongue assembly. In this regard, both the two axle hoop sliders and the two front hoop sliders are telescoping sliders that can be manually extended or shortened to accommodate relatively longer or shorter ATV axle lengths. This axle hoop and front hoop assembly designs allow an individual to quickly and easily adjust the end-to-end length of both assemblies to accept ATVs having relatively longer or shorter front and/or rear axle lengths.

(16) The above-described advantages and other advantages comprising our trailer design will now be described with reference to the Figures, in which FIG. 1 is an illustration of a full trailer assembly **10** having an axle hoop assembly **11**, a tongue assembly **12**, and a front hoop assembly **13**. The axle hoop assembly **11** is removably attached to the proximal end of the tongue assembly **12**, and the front hoop assembly **13** can be removed by sliding it off the tongue assembly. The front hoop assembly is fixedly connected to a carriage assembly **14** which rides on a top surface of the tongue assembly, and which permits the front hoop assembly to be easily moved along substantially the entire length of the tongue assembly. As will be described later with reference to FIGS. 5A and 5B, the front hoop carriage assembly **14** has three slider wheels, with two of the slider wheels riding on and in contact with a top surface of the tongue assembly, and one wheel riding on and in contact with an opposite surface of the tongue assembly. The carriage assembly **14** operates to allow the front hoop assembly to freely move in a direction away from or towards the proximal end of the tongue assembly depending upon whether an ATV is being driven onto or off the trailer assembly respectively. And finally with reference to FIG. 1, the trailer **10** has two ramps **15A** and **15B** that are removably attached to the axle hoop assembly **11** to facilitate driving the ATV onto and off the trailer. As was described earlier, the trailer assembly **10** can be quickly and easily disassembled into several component parts for storage by simply removing clevis pins attaching the axle hoop **11** from the tongue **12**, and one section of the tongue from another section, and by sliding the front hoop **13** off the tongue. Both the axle hoop and the front hoop assemblies can be removed from the tongue assembly, and the tongue assembly can be disassembled into two parts. Further, and as necessary for storage, both the axle hoop and the front hoop assemblies can be easily disassembled.

(17) FIG. 2A is an illustration showing the axle hoop assembly **11** described with reference to FIG. 1. The axle hoop assembly **11** is comprised of two axle hoop tubes **20A** and **20B** both of which are fixedly connected to an axle hoop support tongue **24**. The two axle hoop tubes can be composed of any suitable metallic or non-metallic material (i.e., steel), and can be designed to have a four-sided polygonal shape in cross-section, such as a square. According to one embodiment, the dimension of each side of the square is 2.5 inches wide and the thickness of each side is $\frac{1}{4}$ inch, although the dimensions and thickness can be greater more or thinner depending upon the application. The hoop support tongue can be composed of any suitable metallic or non-metallic material, can be square in cross-section with the dimension of each side being 2.5 inches wide and $\frac{1}{4}$ inches thick. The

assembly **11** has two axle hoop sliders **21A** and **21B** each of which slide into opposite ends of the axle hoop tubes **20A** and **20B** respectively. Both sliders are composed of any suitable metallic material, are square in shape, and the dimension of each slider side being 2 inches with a thickness of $\frac{1}{4}$ inch. Each axle hoop slider **21A** and **21B** has a respective wheel hub **22A** and **22B** connected to it for receiving a wheel (not shown). Each axle hoop slider **21A** and **21B** also has a ramp slider, **23A** and **23B** respectively, that are removably connected to the axle hoop sliders and function to receive a ramp shown with reference to FIG. **1**. The position of each axle hoop slider **21A** and **21B** with respect to the axle tubes **20A** and **20B** is maintained by two wire-lock clevis pins that are placed into holes drilled through each axle hoop tube **20A** and **20B** and which holes register with holes drilled through the axle hoops sliders **21A** and **21B**. According to this embodiment, the distance between an end of one axle hoop to the other axle hoop (i.e., length of the axle) can be made longer or shorter by pulling each of the axle hoop sliders **21A** and **21B** out of or pushing them into the axle tubes **20A** and **20B**, thereby permitting the axle hoop assembly to receive ATVs of longer or shorter axle lengths. In this regard, the axle tube assembly has telescoping functionality.

(18) Continuing to refer to FIG. **2A**, the axle hoop assembly **11** functions to receive rear wheels comprising an ATV. The hoop assembly **11** is designed so that each ATV wheel is securely supported by the hoop assembly **11** in only two places. This hoop design minimizes the metal material needed to support an ATV wheel which has the consequence of minimizing the axle hoop assembly weight, and has the further advantage of allowing the axle hoop assembly to securely support ATVs having a range of wheel diameters [Question . . . any other advantages].

(19) FIG. **2B** is an illustration showing substantially the same axle hoop assembly **11** described earlier with reference to FIG. **2A** with the axle hoop sliders **21A** and **21B** removed from the axle hoop tubes **20A** and **20B**. According to the embodiment shown in FIGS. **2A** and **2B**, the axle hoop sliders are constructed of tubing that is generally in a “U” shape. As can be seen, each axle hoop slider and axle hoop tube have four through holes drilled to receive clevis pins, which when inserted into the axle tube holes and through the holes in the axle hoop slider serve to retain the position of the axle tube sliders with respect to the axle hoop tube. The axle hoop tubes and the axle hoop sliders can have more or fewer holes than shown in the Figure, and the dimensions and gauges of the tubes can be greater or lesser depending upon the type of vehicle being carried.

(20) FIG. **3A** is an illustration showing the tongue assembly **12** described earlier with reference to FIG. **1**. The tongue assembly **12** is designed to serve as a connection between the axle hoop assembly **11** and the slider wheel assembly **14**. The slider wheel assembly is designed to ride on and move along the tongue assembly between a distal hitch ball end **32B** and a proximal axle hoop end **32A**. The tongue assembly **12** is comprised of two, straight tube sections **31A** and **31B** composed of any suitable metallic material, and with each tube being a four-sided polygonal shape in cross section, and each side of the polygon being 3 inches wide and the thickness being $\frac{1}{4}$ inch. The two tongue sections **31A** and **31B** are removably connected to each other by a tongue connector (not shown) that inserts into an open end of each tongue **31A** and **31B**, and the two sections of the tongue assembly are held in place by one or more clevis pins inserted into through holes in the tongue sections and the tongue connector. This design arrangement permits the tongue assembly **12** to be quickly and easily disassembled for storage.

(21) FIG. **3B** is an illustration showing the tongue assembly **12** described with reference to FIG. **3A**. The tongue assembly in FIG. **3B** is exploded to show that it can be disassembled, and to show a tongue connector **34** that when inserted into an end **35** of the tongue **31A** and into an end **36** of the tongue **31B** connects one tongue section to the other tongue section.

(22) FIG. **4A** is an illustration showing the front hoop assembly **13** described earlier with reference to FIG. **1**. The assembly **13** is comprised of front hoop tubes **40A** and **40B**, two front hoop sliders **41A** and **41B**, and a front hoop carriage assembly **14**. The front hoop tubes are parallel with respect to each other, and the front hoop sliders **41A** and **41B** are shown to be slidably connected into opposite ends of the tubes **40A** and **40B** respectively. The hoop sliders **41A** and **41B** are held in

position with respect to the front hoop tubes by clevis pins which are inserted into through holes in the tubes **40A** and **40B** and the sliders **41A** and **41B**. The tubes **40A** and **40B** have a four-sided polygonal shape with the dimension of each side of the polygon being 3 inches in width and the metal thickness being $\frac{1}{4}$ inch, and the sliders **41A** and **41B** also have a four-sided polygonal shape with the dimension of each side of the polygon being 2.5 inches wide and the metal thickness being $\frac{1}{4}$ inch. The carriage assembly **14** can be connected to the front hoop assembly at the mid-points **43A** and **43B** along the lengths of the front hoop tubes **40A** and **40B**. The design and operation of the carriage assembly **14** will be described later in detail with reference to FIGS. **5A** and **5B**. As with the axle hoop assembly **11**, the front hoop assembly **13** is designed so that each ATV wheel is securely supported by the hoop assembly **14** in only two places. This hoop design minimizes the metal material needed to support an ATV wheel which minimizes the axle hoop assembly weight, and has the further advantage of allowing the axle hoop assembly to securely support ATVs having a range of wheel diameters.

(23) FIG. **4B** is an exploded view of the front hoop assembly **13**, described earlier with reference to FIG. **4A**, that serves to illustrate the component parts of the assembly **13**. The slider hoops **41A** and **41B** are inserted into opposite ends of the front hoop tubes **40A** and **40B**, and can be held in place by clevis pins as described earlier with reference to FIG. **4A**. According to this embodiment, the front hoop sliders are shown to in “U” shape. Depending upon an axle length of any particular ATV being carried by the trailer **10**, the end-to-end length of the front hoop assembly **13** can be adjusted by simply removing the clevis pins and moving the sliders **41A** and **41B** into or pulling them out of the front hoop tubes **40A** and **40B** and then replacing the pins. In this regard, the front hoop assembly has telescoping functionality.

(24) FIG. **5A** is an illustration showing the front hoop carriage assembly **14** riding on a portion of a top surface of the tongue assembly **12**. The carriage assembly **14** is comprised of two top C channels **50A** and **50B** between which are retained two slider wheels **51A** and **51B** so that both wheels are in contact with a top surface of the tongue assembly **12**. The position of each top slider wheel between the two top C channels is retained by a bolt that passes into a hole in one of the top C channels, through a slider wheel and out of a hole in an opposite top C channel. The bolts are secured with nuts as shown in FIG. **5**. Further, the carriage assembly **14** is also comprised of two bottom C channels **52A** and **52B** between which is retained one slider wheel **54** so that it is in contact with a bottom surface of the tongue assembly that is opposite to the top surface. As shown in FIG. **5A**, each of the two bottom C channels are fixedly connected to each of the two top C channels by two or more bolts that pass-through holes in the top C channels and through holes in bottom C channels, and being secured by nuts that when tightened securely pull the bottom slider wheel and the top slider wheels into contact with the bottom and top surfaces of the tongue assembly respectively. The bottom slider wheel serves to maintain the top slider wheels in substantially continuous contact with the top surface of the carriage assembly during a time that the front hoop assembly **13** is moved from one position to another along the tongue assembly **13**.

(25) FIG. **5B** is an illustration showing the front hoop carriage assembly **14** described with reference with FIG. **5A** without the portion of the tongue assembly **14**.

(26) FIG. **5C** is an illustration showing the front hoop carriage assembly **14** without the bottom slider wheel **54**. The bottom slider wheel (not shown) rotates on bearings **56A** and **56B** mounted in the wheel, and a bolt **55** through both bottom C channels and the bearings serves to retain the slider wheel in place between the two bottom C channels **52A** and **52B**. The two bearings **56A** and **56B** fit into a bearing cavity (shown with reference to FIG. **5D**) on opposite ends of the bottom wheel **54**.

(27) FIG. **6** is an illustration showing the slider wheel **54** which is cylindrical in shape, can be fabricated out of a solid piece of metallic material, and has a surface **60** that is in contact with the bottom surface of the tongue assembly **11**. The dimension **62** of slider wheel surface **60** should be at least as wide as the surface of the tongue assembly **11**, and in this case, is three to three and one

sixteenth inches wide. The slider wheel has a lip **61** that functions to retain the position of the slider wheel with respect to the bottom surface of the tongue **14** so that it does not wander from side to side with respect to the bottom surface of the tongue (i.e., tracks in a straight line). This lip can be $\frac{1}{4}$ inch high, with respect to the surface **60**, and can be $\frac{1}{4}$ inch thick. The outside diameter (including the lip) of the slider wheel **54** can be two and three-quarters of an inch, and the bearing cavity **63**, referred to earlier, can be machined into both sides of the wheel to an appropriate depth to receive the bearing, and can be one and three-eighth in diameter. A through hole in the wheel **54** can be machined to receive a bolt, such as the bolt **54** shown in FIG. 5C.

(28) The forgoing description, for purposes of explanation, used specific nomenclature to provide a thorough understanding of the invention. However, it will be apparent to one skilled in the art that specific details are not required in order to practice the invention. Thus, the forgoing descriptions of specific embodiments of the invention are presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the invention to the precise forms disclosed; obviously, many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain the principles of the invention and its practical applications, they thereby enable others skilled in the art to best utilize the invention and various embodiments with various modifications as are suited to the particular use contemplated. It is intended that the following claims and their equivalents define the scope of the invention.

Claims

1. An open trailer assembly for carrying a vehicle, comprising: an axle hoop assembly connected to a proximal end of a tongue assembly, and a front hoop assembly connected to the tongue assembly by a carriage assembly and able to ride on the tongue assembly between the axle hoop assembly and a distal end of the tongue assembly; wherein each of one or more front tires of the vehicle come into contact with the front hoop assembly in only two places, and each of one or more rear tires of the vehicle come into contact with the axle hoop assembly in only two places while the vehicle is being carried by the open trailer assembly; and wherein the axle hoop assembly has first and second axle hoop sliders inserted into opposing open ends of first and second axle hoop tubes that are oriented parallel with respect to each other and fixedly connected to a support tongue component; and wherein first and second ramps are removably attached to both the first and second axle hoop sliders respectively.
2. The open trailer assembly of claim 1, wherein the tongue assembly is able to be disconnected from both the front hoop assembly and the axle hoop assembly.
3. The open trailer assembly of claim 2, wherein the front hoop assembly is disassembled from the tongue assembly by sliding it off the tongue assembly.
4. The open trailer assembly of claim 1, wherein the axle hoop assembly is connected to the tongue assembly by one or more clevis pins.
5. The open trailer assembly of claim 1, wherein the front hoop assembly has first and second front hoop sliders inserted into opposing open ends of first and second front hoop tubes that are oriented parallel with respect to each other and held in position by the carriage assembly.
6. The open trailer assembly of claim 5, wherein the end-to-end length of the front hoop assembly is adjustable to be longer or shorter by respectively pulling or pushing one or both of the front hoop sliders away from or into the first and second front hoop tubes.
7. The open trailer assembly of claim 1, wherein the first and second ramps are removably attached to first and second ramp supports that are slidably connected to the first and second axle hoop sliders.
8. The open trailer assembly of claim 1, wherein the end-to-end length of the axle hoop assembly is adjustable to be longer or shorter by respectively pulling or pushing one or both of the axle hoop

sliders away from or into the first and second axle hoop tubes.

9. The open trailer assembly of claim 1, wherein the carriage assembly is comprised of: first and second top C channels between which are retained first and second slider wheels both of which are in contact with a top surface of the tongue assembly; and first and second bottom C channels between which is retained a third slider wheel that is in contact with a bottom surface of the tongue assembly and the first and second bottom C channels are connected to the first and second top C channels respectively.

10. The open trailer assembly of claim 9, wherein both the first and second top C channels comprising the carriage assembly are connected to the front hoop by the first and second front hoop tubes, and the combination of the carriage assembly and front hoop assembly is caused to move from a starting position to a final position, with respect to the tongue assembly, by the one or more front tires comprising the vehicle as it moves onto the open trailer assembly.

11. The open trailer assembly of claim 10, wherein the starting position of the front hoop assembly is proximate to the axle hoop assembly, and the final position of the front hoop assembly is a distance from the axle hoop assembly that is determined by a distance between the front tires and the rear tires of the vehicle.
